Types of Models in Gen AI & their Applications

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Outline

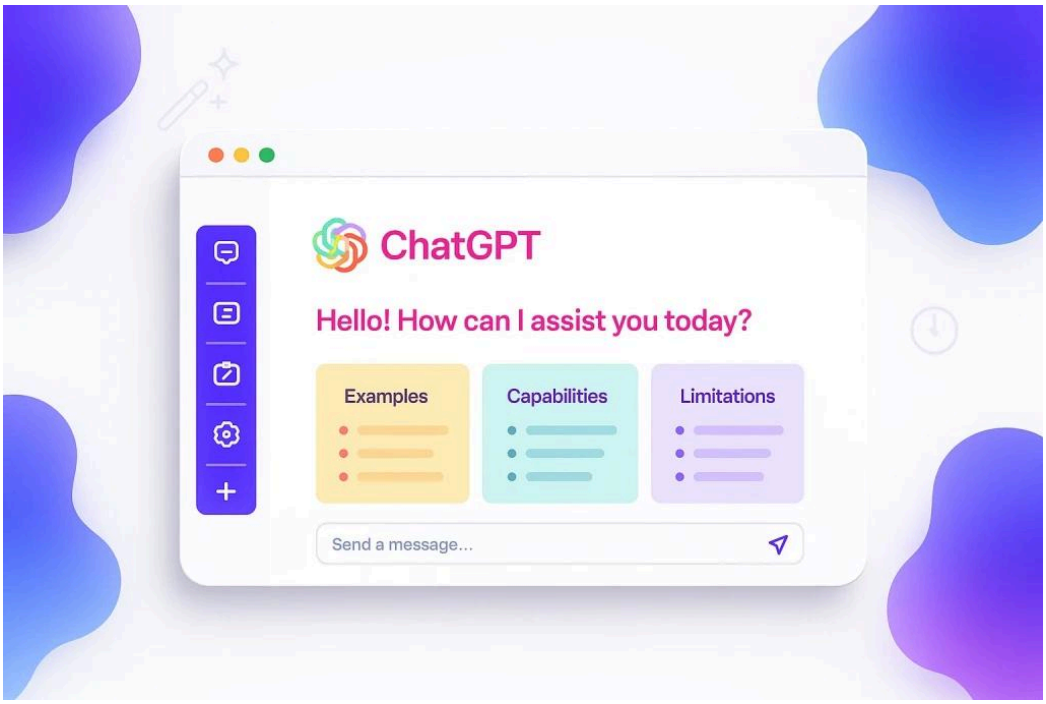
- *Key Models (GPT, Dall-E, Codex, Stable Diffusion)*
- *What is Diffusion and GANs*
- *What are Transformers*
- *Application in Enterprise for AI and their Use cases*

Key AI models

GPT — Generating and understanding text

GPT stands for Generative Pre-trained Transformer. It learns patterns from large amounts of text and generates responses based on the context of a prompt.

- Example: Ask ChatGPT, “Explain machine learning to a beginner,” and it generates an explanation.
- Uses: Summarization, chatbots, report writing, document analysis, and reasoning assistance.



DALL·E — Generating images from text

DALL·E is an image-generation model developed by OpenAI. It turns a written description into a visual image.

- Example prompt: “Create a realistic cat wearing sunglasses on a beach.”
- Uses: Advertising creatives, concept art, product mockups, and visual storytelling.



Codex — Generating and working with code

Codex originated as an OpenAI code model family designed to translate natural-language instructions into programming code. The name is also used for newer OpenAI coding products and agents.

- Example prompt: “Write a Python function to find a number in a sorted array.”
- Uses: Code generation, debugging, testing, explaining code, and software development.



Stable Diffusion — Image generation with diffusion

Stable Diffusion is a family of image-generation models associated with Stability AI. It generates images from prompts using diffusion techniques and supports substantial customization.

- Example prompt: “A futuristic city with flying cars at sunset.”

- Uses: Digital art, design, game assets, marketing images, and image editing.



What are Diffusion Models and GANs?

- A diffusion model learns to generate an image by reversing a process that gradually adds noise to data.

Imagine starting with a clear photograph, adding noise until it becomes static, and then training a model to reverse that process. During generation, the model starts with random noise and progressively creates a meaningful image.



Illustration: random noise → intermediate image → finished landscape.

Example: You enter “A red sports car driving through a rainy Tokyo street at night.” A diffusion model iteratively generates an image that matches your description.

Where it is used: Stable Diffusion, image generation, image editing, visual design, and certain video-generation systems.

- GANs use two neural networks that compete with each other.

The generator improves by learning to fool the discriminator, while the discriminator improves at detecting generated data.

Example: A GAN generates artificial human faces. The discriminator compares generated faces with real faces

from training data, helping the generator produce increasingly realistic results.

Where it is used: Synthetic image creation, image enhancement, face synthesis, and some data-augmentation tasks.

What are Transformers?

A Transformer is a deep-learning architecture introduced in the 2017 research paper Attention Is All You Need. It is the foundation of GPT and many other modern language models.

Its key innovation is self-attention: the model learns which parts of an input are important in relation to other parts.

Enterprise AI: Applications and real-world use cases

Enterprise AI means applying AI within an organization to improve productivity, reduce costs, automate work, and support better decisions.

1. **Customer support and chatbots**

Technologies: GPT, Transformer-based LLMs

A company connects an AI assistant to its FAQs, policies, and order database.

Example: An e-commerce customer asks, “Where is my order?” The assistant retrieves the order status and responds automatically.

Business value: Faster responses, 24/7 support, and reduced workload for human agents.

2. **Document processing and knowledge search**

Technologies: Transformers, embeddings, RAG

AI extracts information from contracts, policies, invoices, and reports.

Example: An employee asks, “What is the procurement approval limit?” A retrieval-augmented generation (RAG) system searches approved internal documents and generates an answer grounded in the retrieved passages.

Business value: Less manual searching and faster access to organizational knowledge.

3. **Software development**

Technologies: GPT, Codex-style coding models

AI helps developers write functions, explain unfamiliar code, generate tests, and debug errors.

Example: A developer asks an AI coding assistant to create a Python API endpoint for retrieving product information.

Business value: Faster development and less repetitive coding, with human review still required.

4. **Marketing and creative content**

Technologies: GPT, DALL·E, Stable Diffusion

AI generates advertising copy, product concepts, campaign images, and social media content.

Example: A shoe company generates several advertising concepts for a new sneaker before selecting one for its campaign.

Business value: Faster creative iteration and lower prototyping costs.

5. **Data analysis and decision-making**

Technologies: Transformers, machine learning, analytics systems

AI can summarize trends, explain reports, and support predictive analytics.

Example: A bank combines machine-learning models with transaction data to flag suspicious activity for investigation.

Business value: Faster insights, improved monitoring, and better-informed decisions.

6. **Healthcare and research**

Technologies: Vision models, Transformers, specialized AI models

AI helps analyze medical images, search scientific literature, and identify potential drug candidates.

Example: A research team uses an AI system to prioritize molecules for further laboratory testing.

Business value: Faster research workflows, with clinical and scientific validation remaining essential.